

PORT ALSWORTH 1 SW, AK, USA

WMO: 703490

Lat: 60.1951N

Lon: 154.3196W

Elev: 321

StdP: 14.53

Time Zone: -9.00 (NAK)

Period: 09-19

WBAN: 26562

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.3	-17.3	-30.5	1.0	-25.1	-22.2	1.6	-16.3	8.7	36.7	8.0	40.1	0.0	N/A	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	76.5	59.8	72.1	57.6	68.4	56.0	61.3	73.5	59.3	69.5	57.5	65.9	3.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.3	68.4	64.3	54.8	64.5	62.0	53.3	61.1	59.9	27.3	73.6	26.0	69.3	24.8	66.3	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	6.1	5.0	DB	-24.8	82.9	9.5	5.0	-31.7	86.5	-37.3	89.4	-42.6	92.2	-49.5	95.9	(4)
(5)			WB	-25.1	63.3	9.4	2.9	-31.9	65.4	-37.4	67.1	-42.7	68.8	-49.6	70.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	17.7	22.4	21.8	35.4	44.9	53.0	56.7	54.4	46.9	36.8	23.3	18.7	(6)
(7)		DBStd	18.17	19.98	14.57	13.83	7.43	5.56	4.76	4.85	4.34	5.73	8.68	13.05	16.59	(7)
(8)		HDD50	5606	1003	774	874	438	174	21	4	10	120	415	802	971	(8)
(9)		HDD65	10567	1468	1194	1339	888	622	361	262	328	542	875	1252	1436	(9)
(10)		CDD50	519	0	0	0	0	17	111	211	147	28	5	0	0	(10)
(11)		CDD65	6	0	0	0	0	0	1	5	0	0	0	0	0	(11)
(12)		CDH74	243	0	0	0	0	9	99	128	7	0	0	0	0	(12)
(13)		CDH80	53	0	0	0	0	0	24	29	0	0	0	0	0	(13)
(14)	Wind	WSAvg	1.8	1.8	2.0	2.0	2.3	2.1	2.0	1.7	1.4	1.5	1.7	1.7	1.9	(14)
(15)	Precipitation	PrecAvg	15.7	1.0	1.0	0.7	0.7	0.5	0.7	1.5	2.2	2.7	1.7	2.3	1.3	(15)
(16)		PrecMax	24.4	2.8	2.7	2.0	1.7	1.4	1.7	4.3	4.3	6.7	3.6	14.3	3.1	(16)
(17)		PrecMin	10.2	0.2	0.1	0.1	0.2	0.1	0.2	0.2	0.8	0.4	0.4	0.2	0.3	(17)
(18)		PrecStd	3.8	0.7	0.7	0.4	0.4	0.4	0.4	0.9	1.1	1.5	0.9	2.9	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	48.0	51.4	56.9	74.4	83.8	84.6	74.5	67.6	59.6	50.6	46.7	(19)
MCWB			42.7	40.2	41.0	42.9	54.6	60.4	65.6	59.8	55.4	47.5	42.2	38.5	(20)	
2%		DB	46.8	44.5	46.9	53.6	68.9	76.2	77.2	70.7	62.7	55.4	48.1	44.6	(21)	
		MCWB	39.3	37.0	38.7	41.7	51.4	58.2	61.5	58.4	52.4	45.2	40.7	37.4	(22)	
5%		DB	44.2	42.1	43.6	51.0	62.8	70.7	72.8	67.1	59.3	52.6	44.8	42.4	(23)	
		MCWB	37.7	35.6	36.9	40.4	48.5	55.8	59.2	56.4	50.8	43.8	39.3	36.4	(24)	
10%		DB	42.0	40.1	40.8	48.5	58.3	66.0	68.7	64.1	56.7	49.9	41.4	39.7	(25)	
		MCWB	36.5	34.4	34.8	39.4	46.4	53.4	57.4	55.4	49.9	42.9	36.9	34.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.0	40.3	41.8	44.4	55.3	62.2	65.9	61.5	55.9	48.0	43.7	39.5	(27)
MCD B			52.1	45.9	49.8	54.4	71.3	79.4	83.2	71.1	64.6	55.5	49.3	44.7	(28)	
2%		WB	40.2	37.4	39.4	42.3	52.1	59.3	62.6	59.6	53.9	46.1	41.5	38.1	(29)	
		MCD B	46.0	43.0	46.5	51.6	66.6	74.7	75.7	67.2	59.8	52.8	46.7	43.2	(30)	
5%		WB	38.3	36.0	37.5	41.1	49.3	56.6	60.3	58.1	52.4	44.7	39.7	36.9	(31)	
		MCD B	43.5	41.2	43.3	49.9	61.4	69.1	71.2	65.1	57.3	50.7	44.5	41.7	(32)	
10%		WB	36.5	34.7	35.3	39.7	47.1	54.6	58.5	56.4	51.0	43.3	37.7	35.1	(33)	
		MCD B	41.7	39.4	40.3	47.5	57.0	64.8	67.7	62.6	55.4	48.5	41.9	39.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.8	16.9	22.1	20.7	23.1	22.9	18.6	16.2	16.9	14.8	13.7	13.7	(35)
MCD BR			13.0	13.6	19.3	25.2	34.4	34.0	29.2	23.5	20.9	17.1	13.7	12.1	(36)	
MCWBR			8.1	8.7	12.3	14.9	18.2	16.7	14.3	12.3	12.3	10.6	8.9	8.0	(37)	
5% WB		MCD BR	12.9	13.2	17.4	22.2	31.2	30.2	26.9	20.2	18.5	14.6	12.6	11.6	(38)	
		MCWBR	8.3	8.8	10.9	13.2	16.9	15.4	13.5	10.7	11.7	9.4	8.6	8.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.240	0.276	0.300	0.390	0.383	0.392	0.370	0.361	0.327	0.299	0.254	0.235	(40)	
taud		2.096	2.214	2.183	1.970	2.167	2.273	2.413	2.424	2.495	2.350	2.060	1.929	(41)		
Ebn at Noon		167	230	261	251	266	266	270	263	255	220	152	106	(42)		
Edh at Noon		14	22	31	47	42	38	33	30	23	19	14	10	(43)		
(44)	All-Sky Solar Radiation	RadAvg	115	326	869	1355	1565	1662	1396	1103	751	370	149	64	(44)	
(45)		RadStd	16	52	115	143	179	139	129	145	112	39	24	7	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+1.49	+4.79	+4.88	N/A	N/A	N/A	N/A	-547	N/A	N/A

Nomenclature: See separate page